

UNIVERSITY EXAMINATION BOARD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
AUTONOMOUS DEGREE SEMESTER EXAMINATION

Subject: Design and Analysis of Algorithms	Sub Code: CS-501	Max Marks: 35
Time / Duration: 3 Hours	Academic Year: 2026-2027	Semester: V Semester

General Instructions:

- 1. Answer all questions.
- 2. Assume suitable data if necessary.

SECTION I

ANALYTICAL & DESIGN PROBLEMS

1. Illustrate the step-by-step mechanism of Introduction: Algorithm using a concrete example. **[2 Marks]**
2. Compare and contrast the advantages and limitations of Disjoint Sets: Disjoint set operations. **[2 Marks]**
3. Explain the fundamental working principle of Dynamic Programming: General method with suitable illustrations. **[2 Marks]**
4. Explain the mathematical or logical formulation underlying Greedy method: General method. **[2 Marks]**
5. Illustrate the step-by-step mechanism of Branch and Bound: General method using a concrete example. **[2 Marks]**

SECTION 2

6. (a) How is Performance Analysis-Space complexity implemented in modern software systems? Explain with pseudocode or schema. **[5 Marks]**

[OR]

- (b) Explain the end-to-end design and implementation of Introduction: Algorithm. Trace its execution on a non-trivial benchmark dataset. **[5 Marks]**
7. (a) Discuss how union and find algorithms handles edge cases, boundary conditions, or exceptions. **[5 Marks]**

[OR]

- (b) Provide a comprehensive, in-depth analysis of Disjoint Sets: Disjoint set operations. Discuss its algorithmic or theoretical design in detail. **[5 Marks]**
8. (a) Illustrate the step-by-step mechanism of applications- Optimal binary search tree using a concrete example. **[5 Marks]**

[OR]

- (b) Design a robust solution using Dynamic Programming: General method for an enterprise-scale engineering problem. Analyze its complexity. **[5 Marks]**

- 9. (a)** Describe the architectural or structural flow involved in applications- Job sequencing with deadlines. **[5 Marks]**

[OR]

- (b)** Design a robust solution using Greedy method: General method for an enterprise-scale engineering problem. Analyze its complexity. **[5 Marks]**

- 10. (a)** Describe the architectural or structural flow involved in applications. **[5 Marks]**

[OR]

- (b)** Design a robust solution using Branch and Bound: General method for an enterprise-scale engineering problem. Analyze its complexity. **[5 Marks]**